Linear and Quadratic Inequalities

Concept

An inequality describes a set of values, often one or more intervals.

Reverse sign

Reverse $<$ or $>$ only when multiplying/dividing by a negative value.

Sign chart

Find zeros, split the number line, test each interval.

Endpoints

$\leq$ or $\geq$ includes zeros; $<$ or $>$ excludes them.

Notation

(a,b) excludes endpoints; $[a,b]$ includes them; $\cup$ joins intervals.

Example 1

Solve $-3x + 7 \leq 16$.

Solution

1. $-3x \leq 9$.
2. Divide by -3 and reverse: $x \geq -3 \rightarrow [-3,\infty)$.

Linear and Quadratic Inequalities

Example 2

Solve $x^2 - 5x + 6 > 0$.

Solution

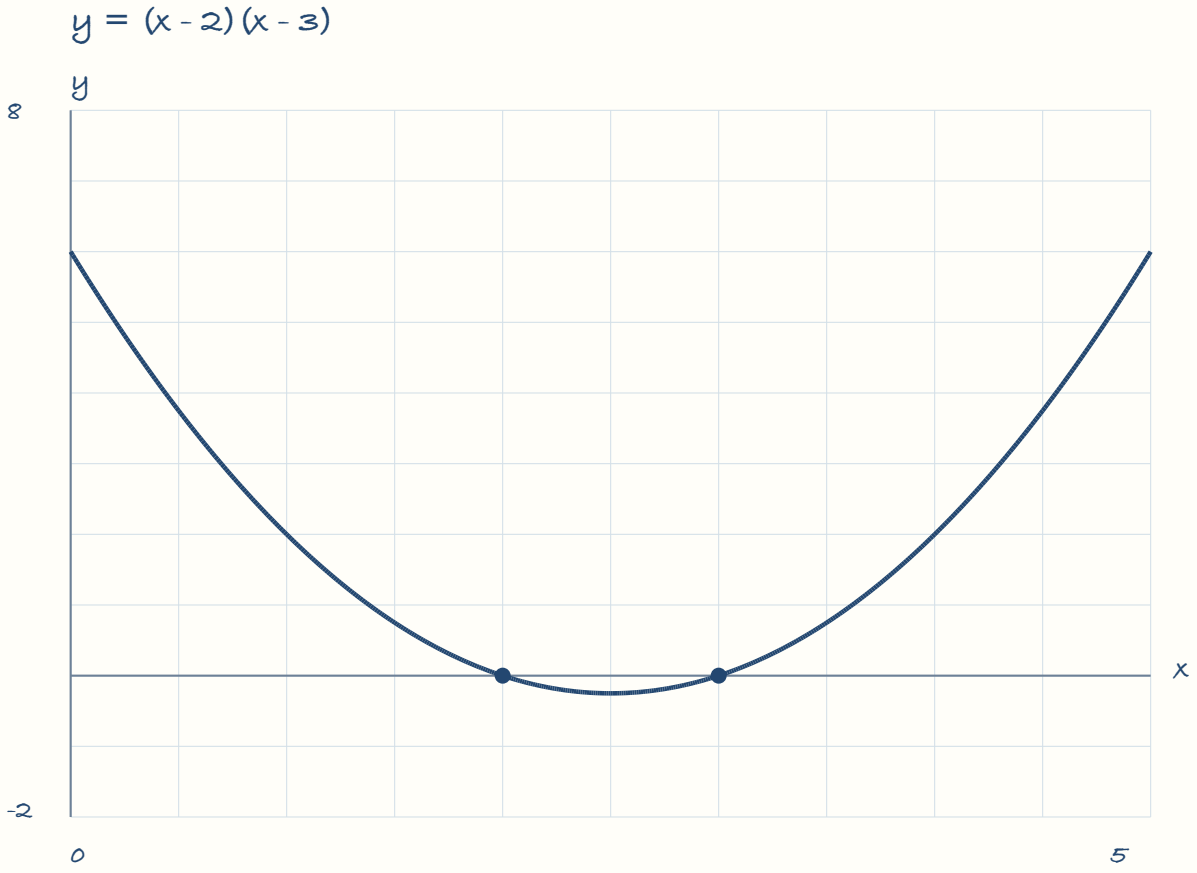
1. Factor: $(x - 2)(x - 3) > 0$. Critical values: 2,3.

2. Product signs: positive for $x < 2$; negative for $2 < x < 3$; positive for $x > 3$.

3. Answer: $(-\infty,2) \cup (3,\infty)$. Exclude the zeros.

Linear and Quadratic Inequalities

Visual notes | read the graph



Positive outside the zeros and negative between them. Strict inequalities exclude the zeros.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Linear and Quadratic Inequalities

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $5 - 2x > 11$.

2. Solve $x^2 - 4 \leq 0$.

3. Solve $(x - 1)(x + 3) \geq 0$.

4. Solve $-(x - 2)^2 > 0$.

Linear and Quadratic Inequalities

Practice answers | check after trying

1. $-2x > 6$ gives $x < -3$.

2. Zeros -2 and 2; the quadratic is non-positive between them: $[-2,2]$.

3. Positive outside the roots and zero at them: $(-\infty,-3] \cup [1,\infty)$.

4. No solution. A square is non-negative, so its negative cannot be strictly positive.